

APIA WHARF, Samoa

WMO: 917560

Lat: 13.826S

Lon: 171.761W

Elev: 10

StdP: 14.69

Time Zone: 13.00 (PCS)

Period: 98-11

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 71.2	(c) 72.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 18.9	(k) 81.1	(l) 17.3	(m) 81.1	(n) 3.1	(o) 180	(p) 0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 3	(b) 9.8	(c) 88.1	(d) N/A	(e) 87.0	(f) N/A	(g) 86.3	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 6.1	(p) 70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.6	(b) 15.6	(c) 13.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.8	80.6	80.7	81.0	80.7	79.8	79.2	78.4	78.6	79.0	79.5	80.2	80.4	
	DBStd	1.61	1.53	1.55	1.57	1.18	1.41	1.46	1.31	1.26	1.19	1.22	1.32	1.37	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD50	10888	947	860	961	920	923	877	881	885	869	915	906	944	
	CDD65	5413	482	440	496	470	458	427	416	420	419	450	456	479	
	CDH74	50851	4738	4356	4946	4681	4205	3822	3419	3600	3761	4204	4413	4706	
	CDH80	12020	1252	1185	1419	1292	975	810	556	599	670	892	1129	1241	
Wind	WSAvg	5.4	5.1	4.8	4.5	4.9	4.9	5.7	5.8	6.4	6.6	5.8	5.0	4.9	
Precipitation	PrecAvg	117.0	17.3	13.0	14.1	9.1	7.0	5.9	4.5	4.9	6.5	9.1	10.4	14.6	
	PrecMax	160.4	37.8	36.8	27.4	20.8	14.4	12.9	11.5	12.1	23.3	22.8	28.6	36.7	
	PrecMin	74.8	9.0	3.1	5.2	0.6	0.4	1.4	0.2	0.7	0.0	1.9	1.0	2.5	
	PrecStd	25.1	7.5	6.1	5.7	4.9	3.2	3.1	2.6	3.1	5.0	6.0	6.1	8.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	88.4	92.5	88.0	86.7	85.5	84.6	84.6	85.0	85.8	88.9	89.4	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	86.8	87.2	88.1	86.5	85.8	84.9	83.9	83.9	84.3	85.1	86.6	86.9	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	86.0	86.3	86.4	86.0	85.2	84.3	83.4	83.4	83.6	84.4	85.5	86.0	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	85.2	85.4	85.7	85.3	84.5	83.7	82.8	82.8	83.1	83.8	84.7	85.1	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2%		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
5%		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
10%		WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.5	9.8	9.5	9.3	9.0	8.8	8.7	8.5	8.7	9.5	9.5	
		MCDBR	10.1	10.4	11.3	10.1	9.9	9.4	9.4	9.3	9.1	9.4	10.4	10.7	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Clear-Sky Solar Irradiance	taub	0.387	0.378	0.375	0.382	0.390	0.391	0.392	0.396	0.399	0.388	0.385	0.385		
	taud	2.625	2.659	2.670	2.644	2.604	2.586	2.573	2.556	2.552	2.597	2.618	2.626		
	Ebn at Noon	303	304	298	286	273	268	270	278	287	297	302	304		
	Edh at Noon	32	31	30	29	29	28	29	31	33	33	32	32		
All-Sky Solar Radiation	RadAvg	1812	1811	1816	1633	1440	1376	1423	1649	1845	1904	1867	1843		
	RadStd	181	169	135	134	102	69	57	78	104	96	98	115		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page